

Homework 1

Section 1.2

(8) Using induction, prove that $7^n - 2^n$ is divisible by 5 for every $n \in \mathbb{N}$.

(9) Using induction, prove that $\sum_{k=1}^n k = \frac{n(n+1)}{2}$.

(13) Let a sequence $\{x_n\}$ of numbers be defined recursively by

$$x_1 = 1 \quad \text{and} \quad x_{n+1} = \frac{1}{1 + x_n}.$$

Prove by induction that x_{n+2} is between x_n and x_{n+1} for each $n \in \mathbb{N}$.

Section 1.3

The two problems in this section use the following definition.

Definition 1.3.1: A *commutative ring* is a set R with two binary operations, *addition* $(a, b) \mapsto a + b$ and *multiplication* $(a, b) \mapsto ab$, that satisfy the following axioms:

- A1.** (Commutative Law of Addition) $x + y = y + x$ for all $x, y \in R$.
- A2.** (Associative Law of Addition) $x + (y + z) = (x + y) + z$ for all $x, y, z \in R$.
- A3.** (Additive Identity) There is an element $0 \in R$ such that $0 + x = x$ for all $x \in R$.
- A4.** (Additive Inverses) For each $x \in R$, there is an element $-x \in R$ such that $x + (-x) = 0$.
- M1.** (Commutative Law of Multiplication) $xy = yx$ for all $x, y \in R$.
- M2.** (Associative Law of Multiplication) $x(yz) = (xy)z$ for all $x, y, z \in R$.
- M3.** (Multiplicative Identity) There is an element $1 \in R$ such that $1 \neq 0$ and $1x = x$ for all $x \in R$.
- D.** (Distributive Law) $x(y + z) = xy + xz$ for all $x, y, z \in R$.

(3) Prove that if $\mathbb{Z}$ satisfies the axioms for a commutative ring, then $\mathbb{Q}$ satisfies **A1** and **M1**.

(4) Prove that if $\mathbb{Z}$ satisfies the axioms for a commutative ring, then $\mathbb{Q}$ satisfies **A2** and **M2**.